

Mini AI Project – End-to-End Machine Learning Workflow

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Internship: AI/ML Internship

Project: Titanic Survival Prediction

1. Problem Statement

The objective of this project is to build a simple end-to-end machine learning model that predicts whether a passenger survived the Titanic disaster based on the available passenger information.

The project demonstrates the complete machine learning workflow, including data preprocessing, exploratory data analysis, model building, and evaluation.

2. Approach

The following steps were performed during the project:

- Loaded the Titanic dataset using Pandas.
 - Explored the dataset to understand its structure and missing values.
 - Filled missing values in the Age and Embarked columns.
 - Removed the Cabin column because it contained many missing values.
 - Converted categorical features (Sex and Embarked) into numerical values.
 - Selected the required features for model training.
 - Split the dataset into training and testing sets.
 - Trained a Logistic Regression model.
 - Evaluated the model using accuracy score.
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3. Results

The Logistic Regression model was successfully trained and tested on the Titanic dataset.

Algorithm Used:

- Logistic Regression

Evaluation Metric:

- Accuracy Score

Model Accuracy:

- (Write your accuracy here, for example: 0.80 or 80%)
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4. Insights

- Data preprocessing improved the quality of the dataset.
 - Logistic Regression performed well for this binary classification problem.
 - Feature selection helped simplify the dataset before training.
 - The project demonstrated the complete workflow of a beginner-level machine learning project from data preparation to model evaluation.
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5. Conclusion

This project successfully demonstrated an end-to-end machine learning workflow using the Titanic dataset. It included data cleaning, preprocessing, feature selection, model training, and evaluation using Logistic Regression. The project provided practical experience in building a machine learning model using Python and Scikit-learn.